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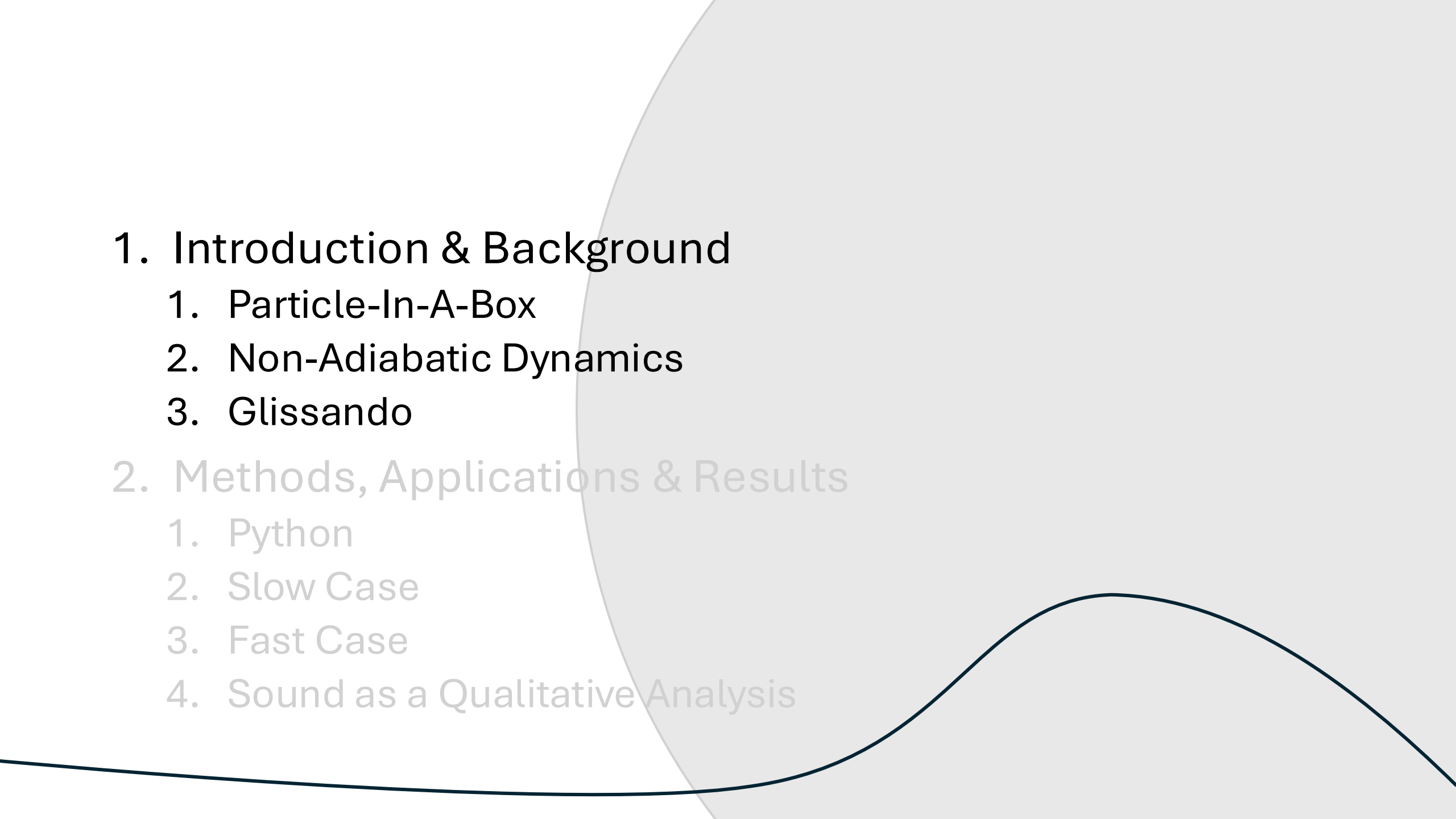
The Quantum Glissando Effect: Expanding the Particle-in-a-Box Model to Include Nonadiabatic Effects



Liz Stippell

She/her/hers

Prof. Abraham Nitzan Group, University of Pennsylvania



1. Introduction & Background

1. Particle-In-A-Box
2. Non-Adiabatic Dynamics
3. Glissando

2. Methods, Applications & Results

1. Python
2. Slow Case
3. Fast Case
4. Sound as a Qualitative Analysis

Quantum vs. Classical Mechanics

Some phenomena cannot be described by classical mechanics alone

- **Probabilistic:** Outcomes are described by probabilities
- Energy is **not** continuous (only certain *quanta* of energy can exist)
- Particles also have a wave form

Schrödinger Equation

$$H\Psi = E\Psi \rightarrow i\hbar \frac{\delta\Psi}{\delta t} = H\Psi$$

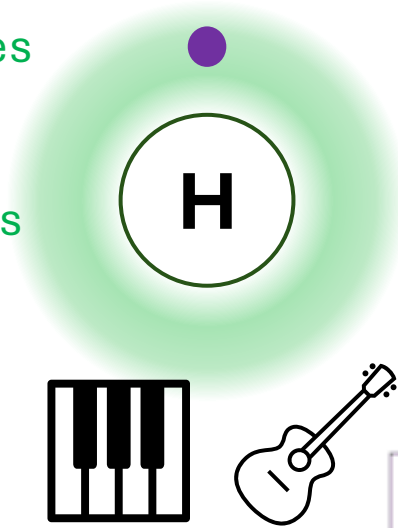
Total energy of a system (K.E. + P.E)

$$H = \frac{-\hbar^2}{2m} \frac{\delta^2}{\delta x^2} + V(x)$$

Wave Function: Ψ

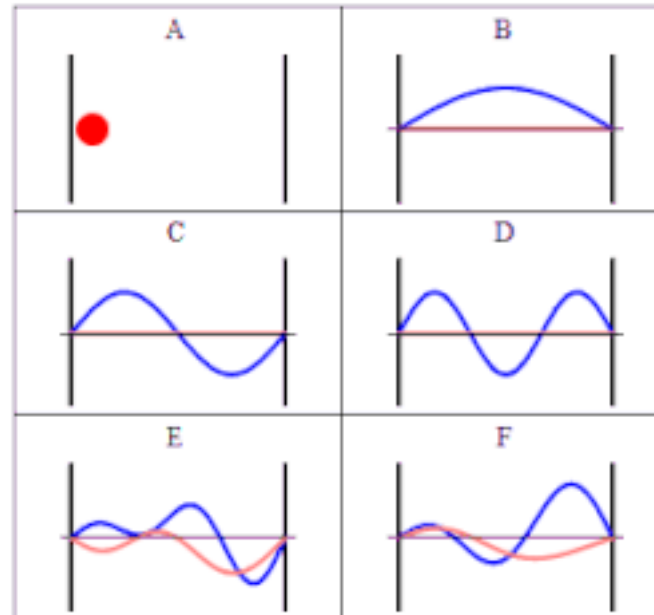
Probabilities: Ψ^2

Energy (Eigen)States

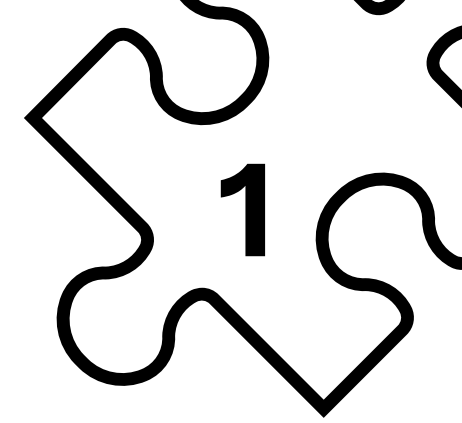


- **Deterministic:** If you know initial conditions you can figure out any future state
- Energy is **continuous**
- Particles are just particles

$$\Psi = \sin(nx)$$



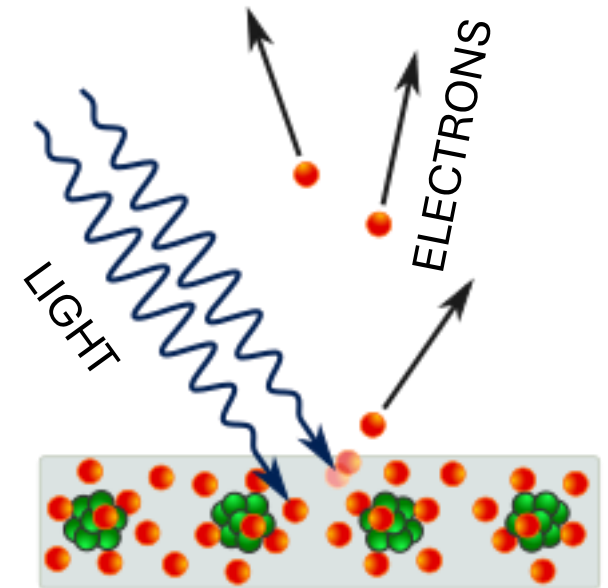
Combinations of states!



The Photoelectric Effect

Albert Einstein's Only Nobel Prize

This effect is how solar panels work!



The Particle-In-A-Box Model

What happens if we put a particle in a one-dimensional box of length L ?

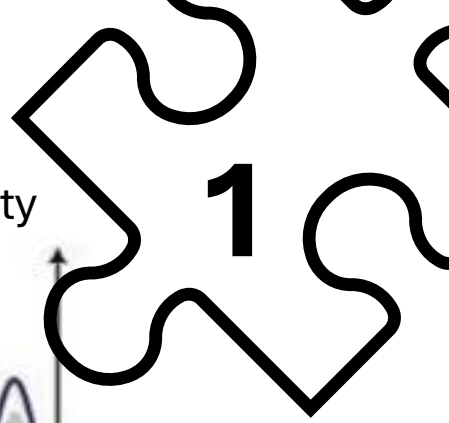
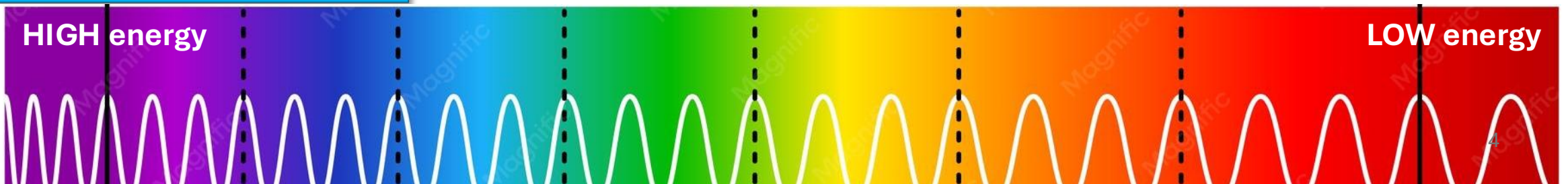
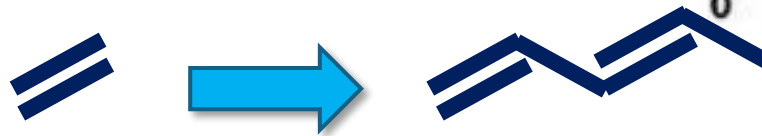
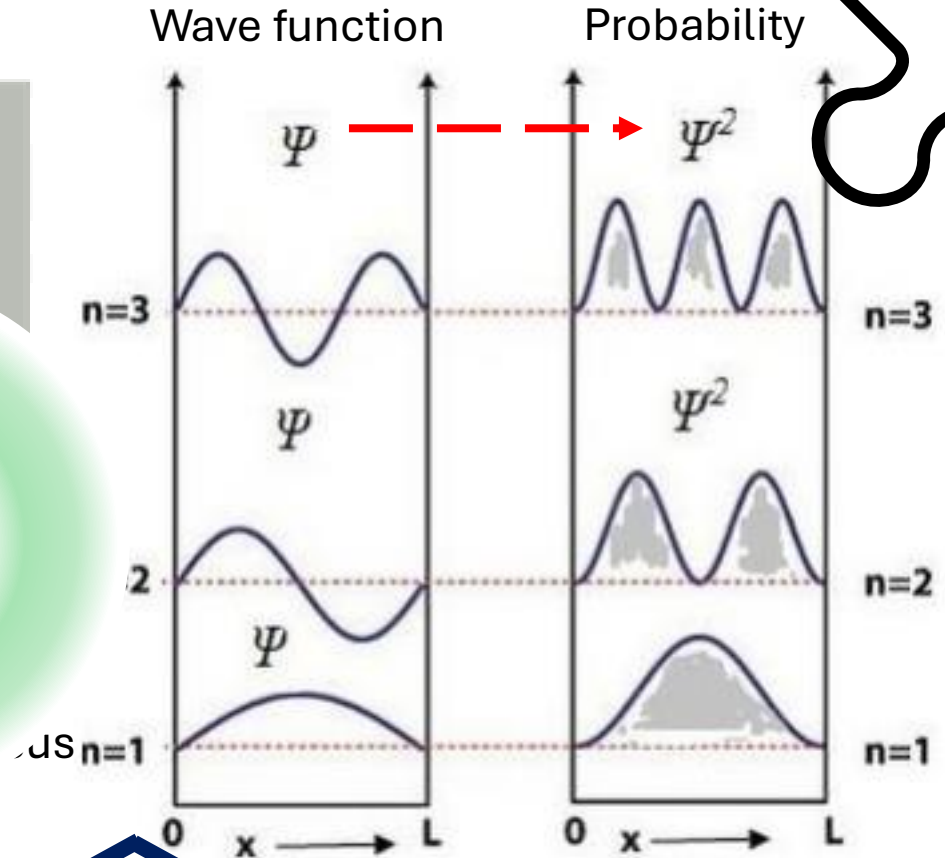
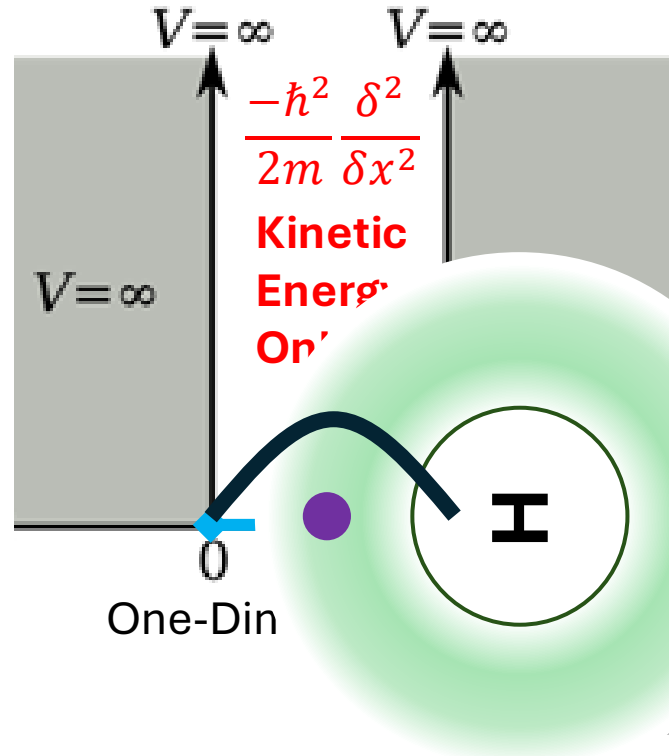
$$H\Psi = E\Psi$$

$$\Psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)$$

$$E = n^2 \frac{h^2}{8mL^2}$$

Quantum number
Length of box

Real-Life Particle in a box:
conjugated carbon chains

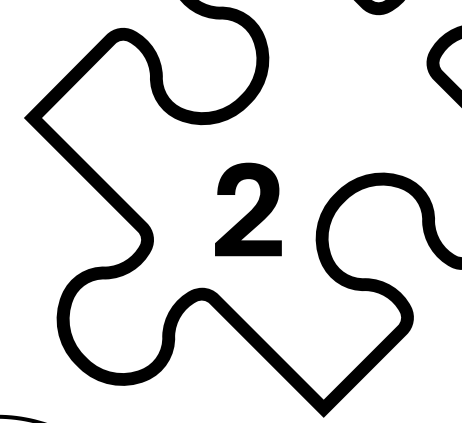


Non-Adiabatic Dynamics

Quantum calculations are **hard!**

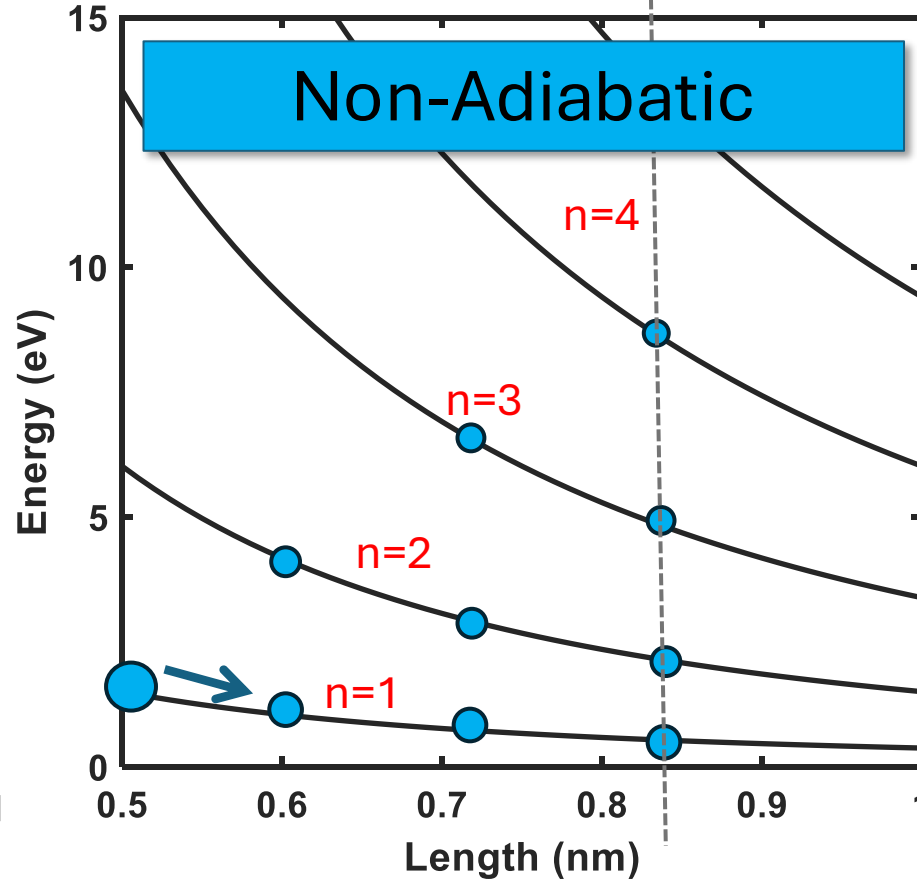
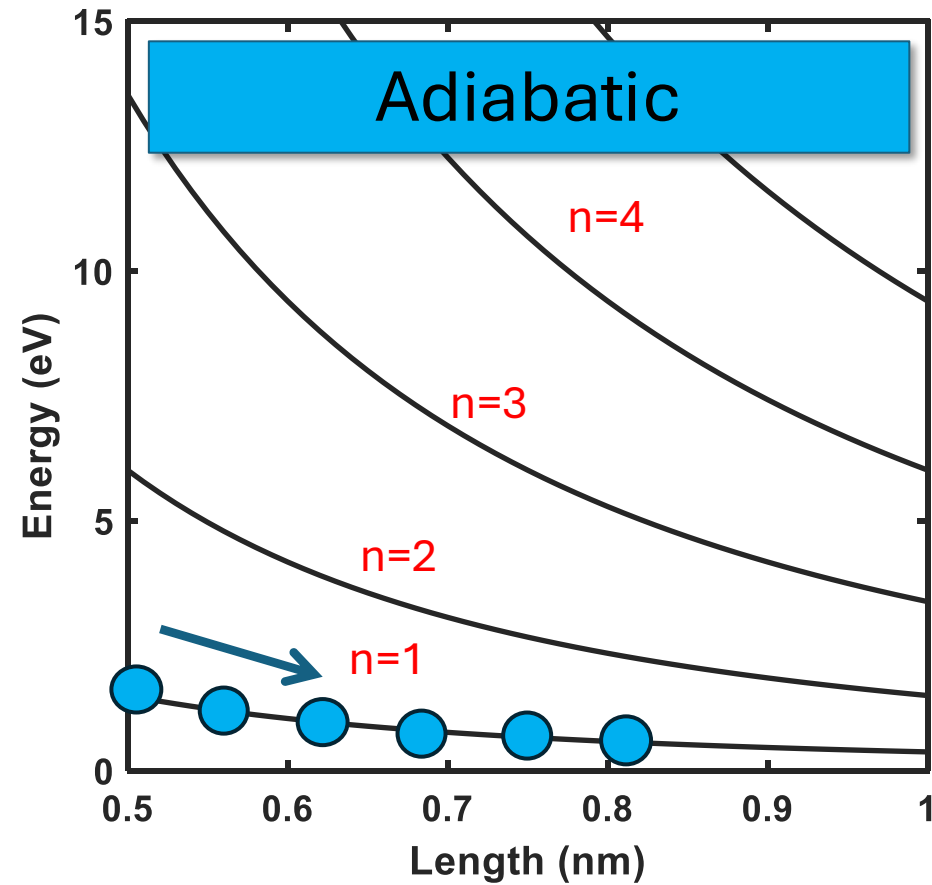
Adiabatic Approximation: To simplify things, we can assume the nucleus moves *a lot* slower than the electrons

Non-Adiabatic: The nucleus' movement is **non-negligible**.



constant

$$E = n^2 \frac{h^2}{8mL^2} = \frac{n^2}{L^2}$$
$$E(n = 1, L = 1) = 1$$
$$E(n = 1, L = 2) = \frac{1}{4}$$
$$E(n = 2, L = 2) = 1$$



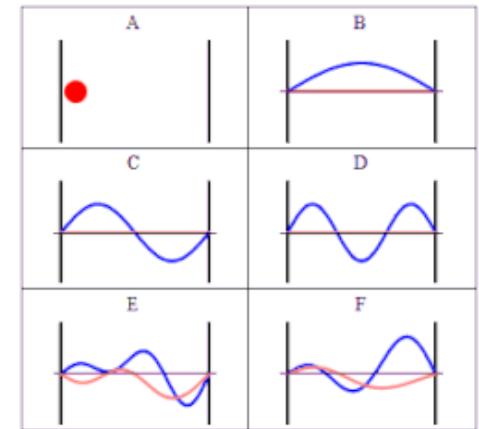
MAIN IDEA:

If we expand our box, energy states get closer together. The faster we expand, the easier it is for particles to mix between states

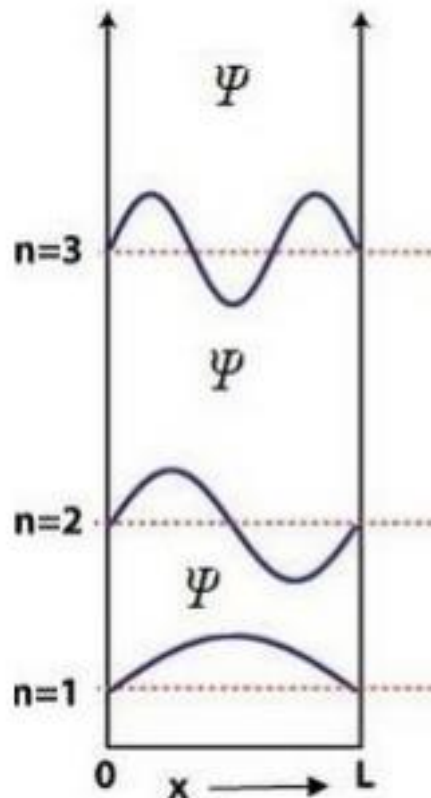
How do we mix states?

Quantum mechanics states energy only exists at **specific** states

- *But* we can add these states together



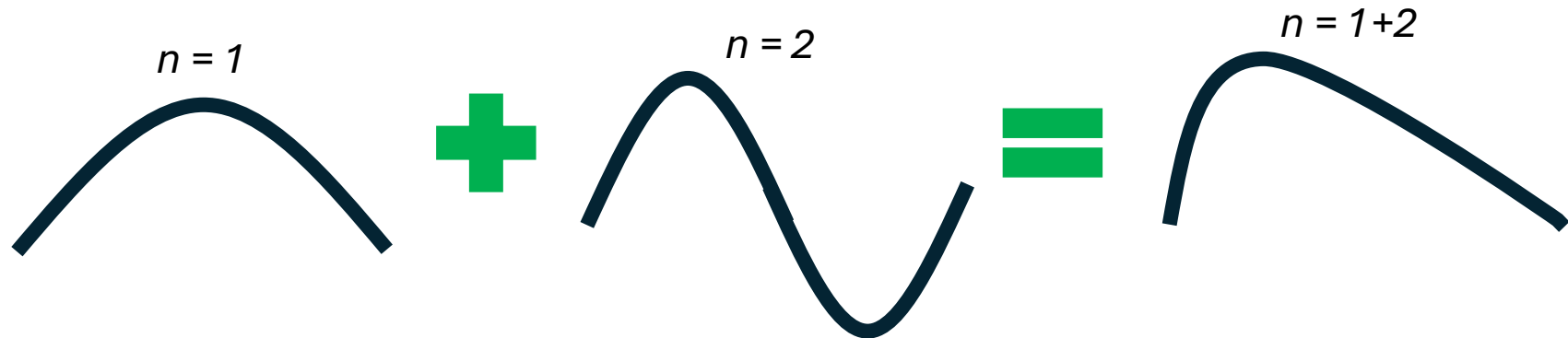
Wave function



$$\Psi = \sum c_n \psi_n = c_1 \psi_1 + c_2 \psi_2 + c_3 \psi_3 + \dots$$

Ground state + 1st Excited + 2nd Excited + ...

Coefficients: Determine “how much” of the particle is in this state (probability – *must add up to one!*)



Constructive (additive) and destructive (subtractive) interference

Superpositions!

How do we find the coefficients?

- **Probabilities** describe the degree of mixing between states, and are determined by the **coefficients**
- The faster the Hamiltonian changes (which describes total energy) and the closer together the energy states are, the more mixing occurs

$$\frac{d}{dt} c_j(t) =$$

The rate of change of
the j^{th} coefficient...

$$\sum_{n \neq j} c_n(t) \text{ ————— }$$

... plus some extra
bits that depend on
other coefficients.

Adiabatic approximation

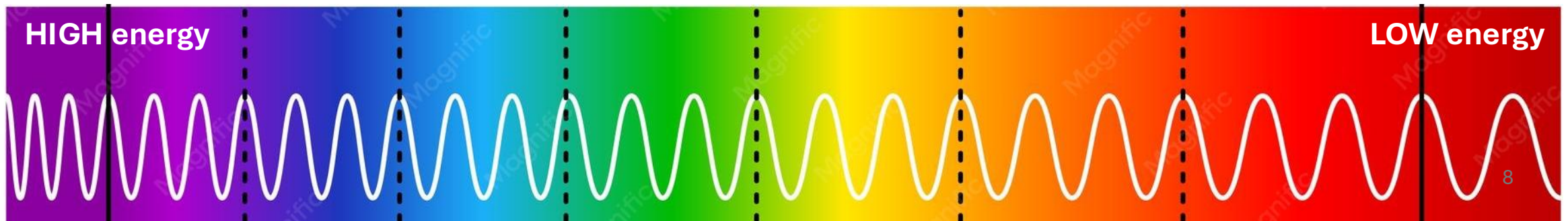
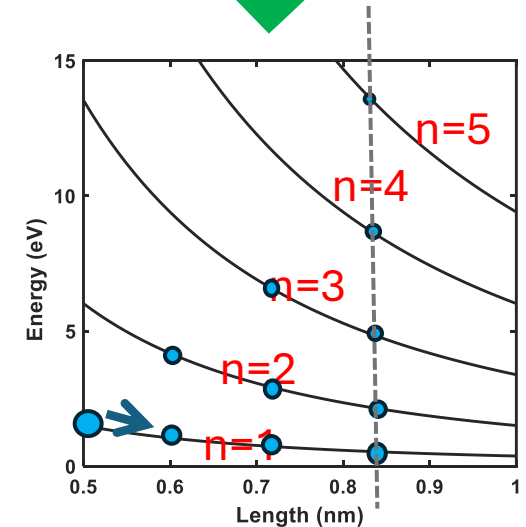
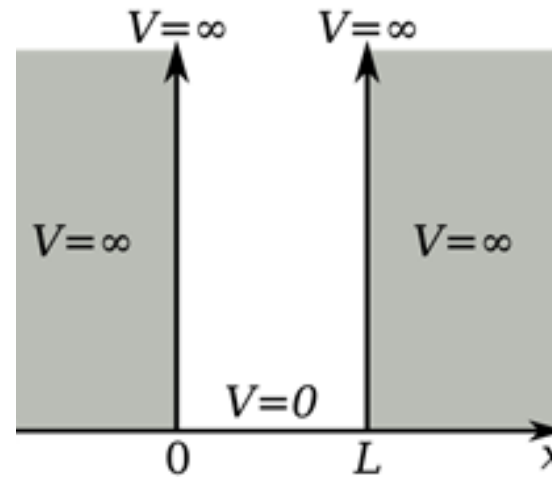
1. Mixture with other states is **directly proportional** to how **fast the Hamiltonian is changing**
2. Mixture of state j with other states inversely depends on **how close they are in energy** to state j .

Introduction: Summary

The Quantum Glissando Effect: Expanding the Particle-in-a-Box Model to Include Nonadiabatic Effects

- **Probabilistic:** Outcomes are described by probabilities
- Energy is **not** continuous (only certain *quanta* of energy can exist)

What is a Glissando?



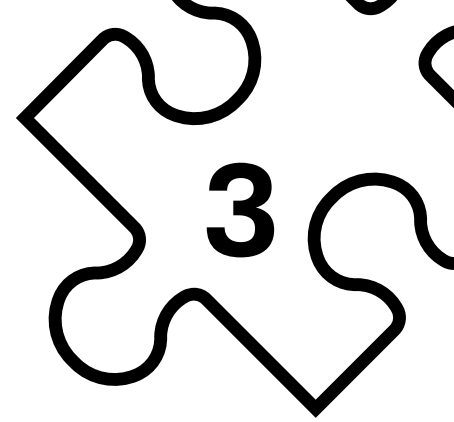
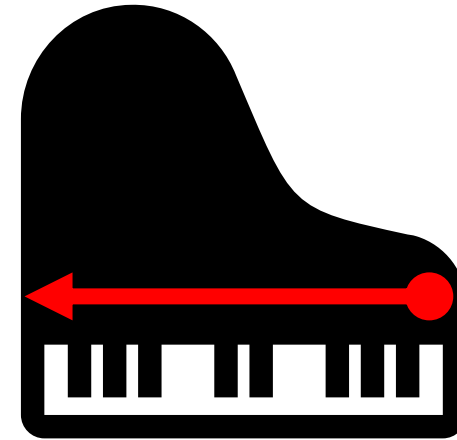
Music Theory: What Is a Glissando?

Glissando.

A glide from one pitch to another.
This can be induced in various instruments.

Guitar: Slide your finger down a string for a *continuous* change in pitch

Piano: Slide your finger down the keys for a *stepwise* change in pitch





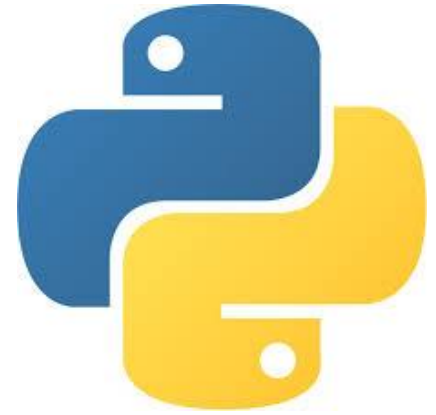
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Python: A versatile tool for any skill level



Free (accessible)
Customizable



ANACONDA®



Two Cases: Slow vs. Fast

Slow

Adiabatic Case

Particle can “keep up” with the expanding box, allowing it to remain almost fully in the ground state

Fast

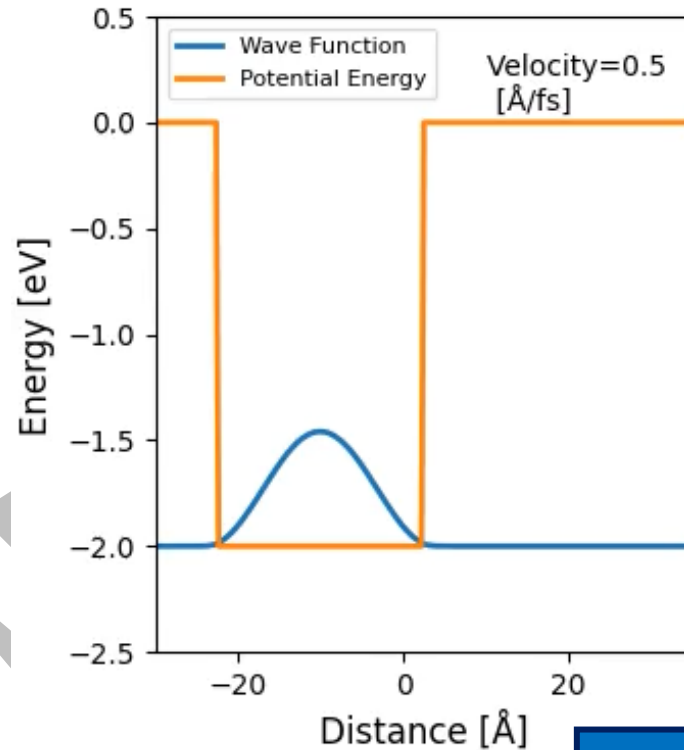
Non-Adiabatic Case

Particle can’t “keep up” with the expanding box, facilitating energy crossings. Couplings with excited states create a wave function which combines multiple energy levels

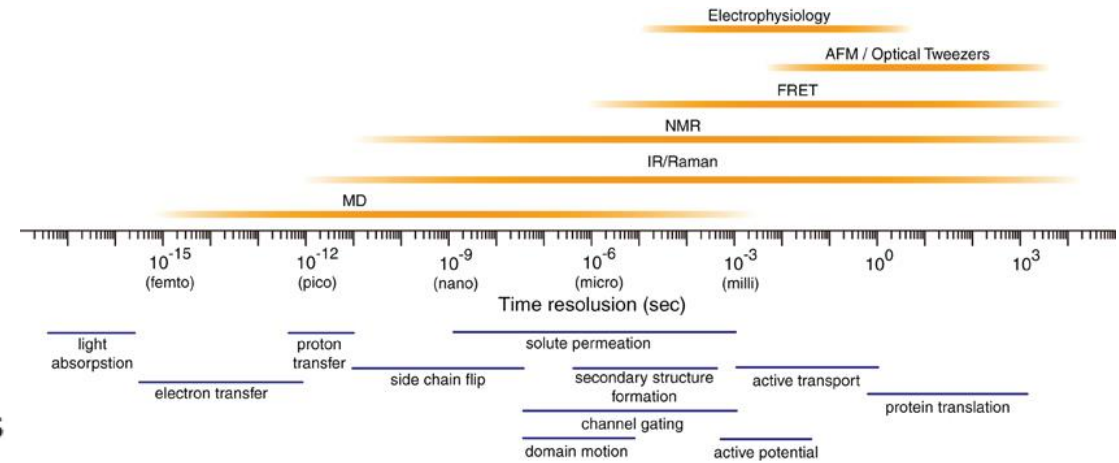
Case One: Slow Expansion

1 femtosecond = 10^{-15} second

Time: 0.00 fs



Size of points
proportional to
contribution



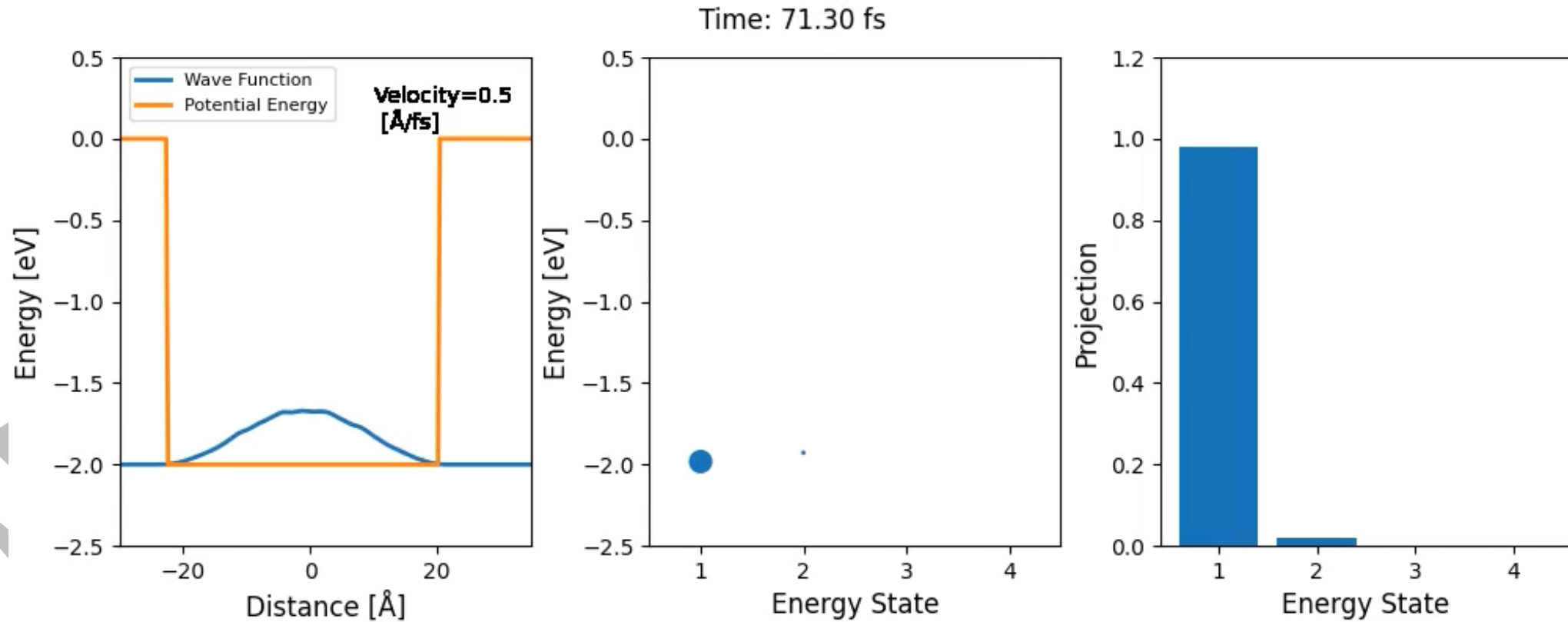
Ground state
 $n = 1$

Case One: Slow Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 0.5 Å/fs

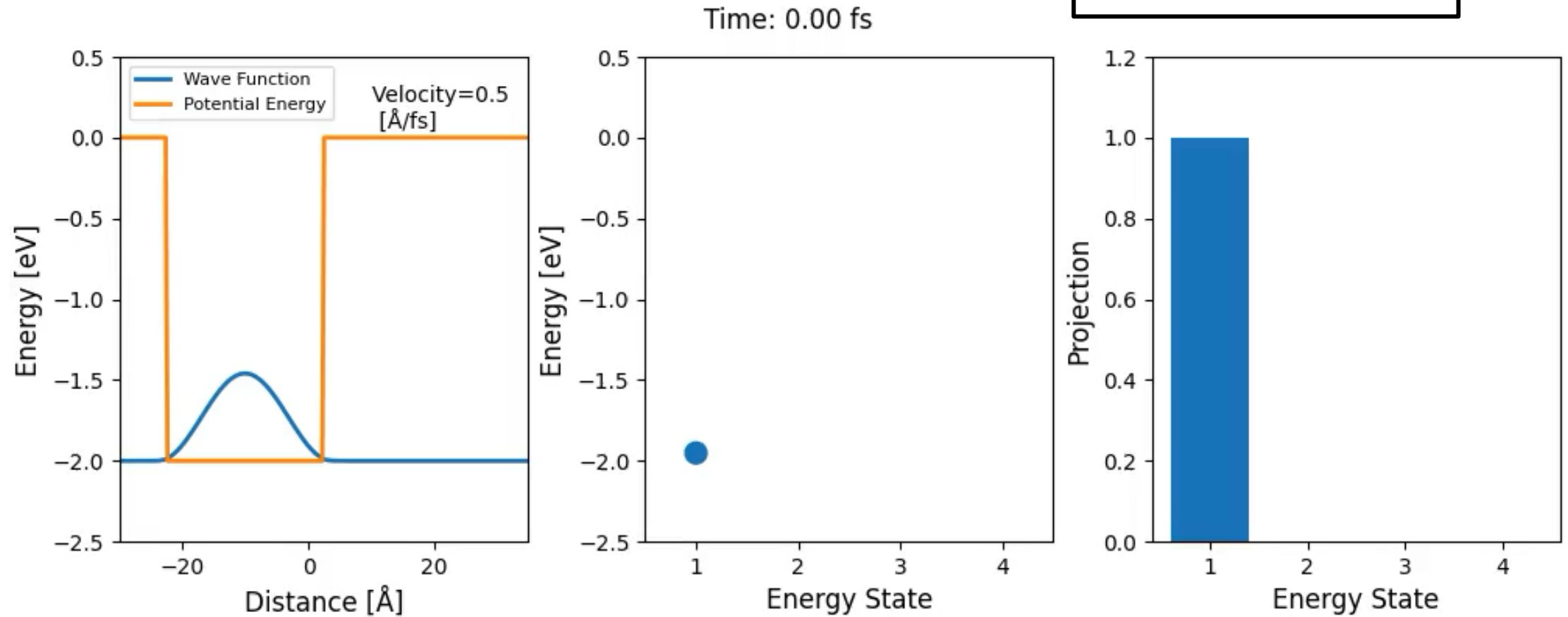


Case One: Slow Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 0.5 Å/fs

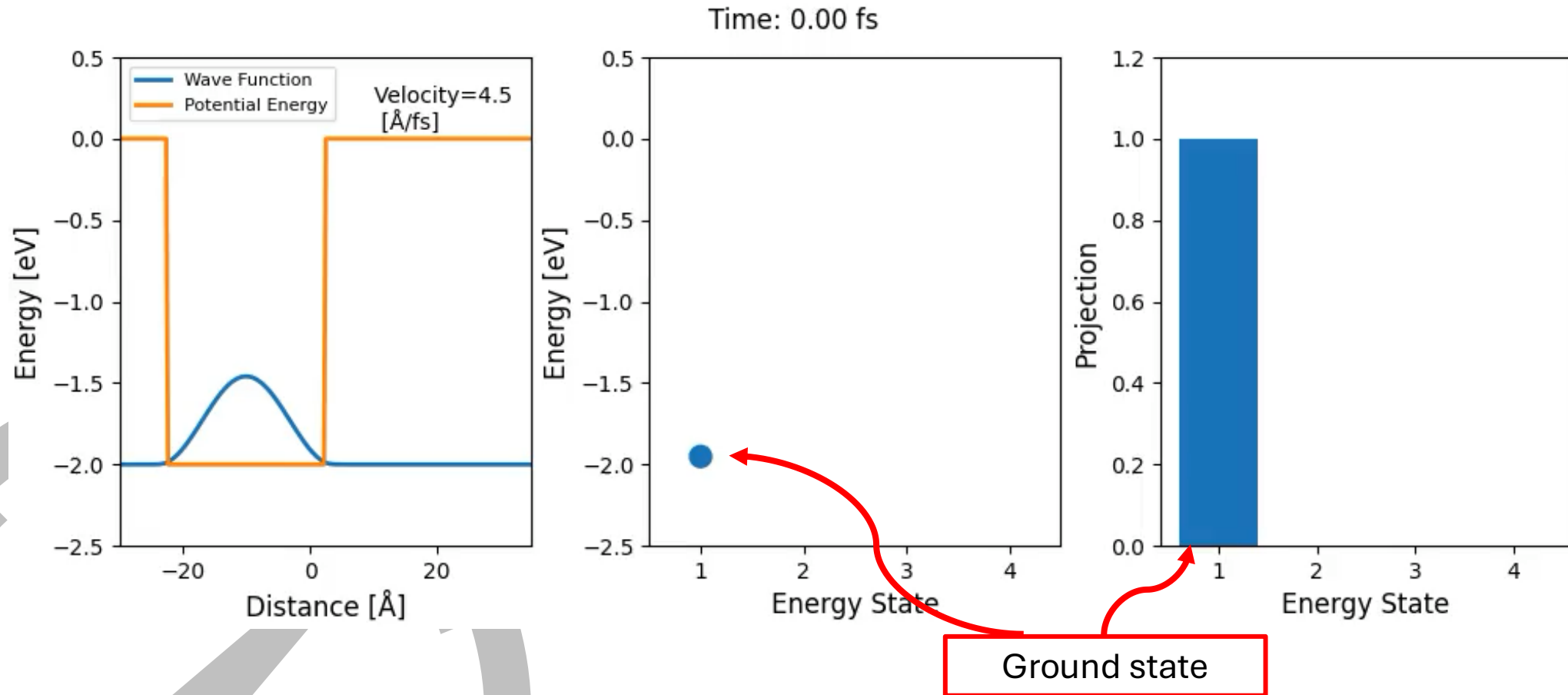


Case Two: Fast Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 4.5 Å/fs

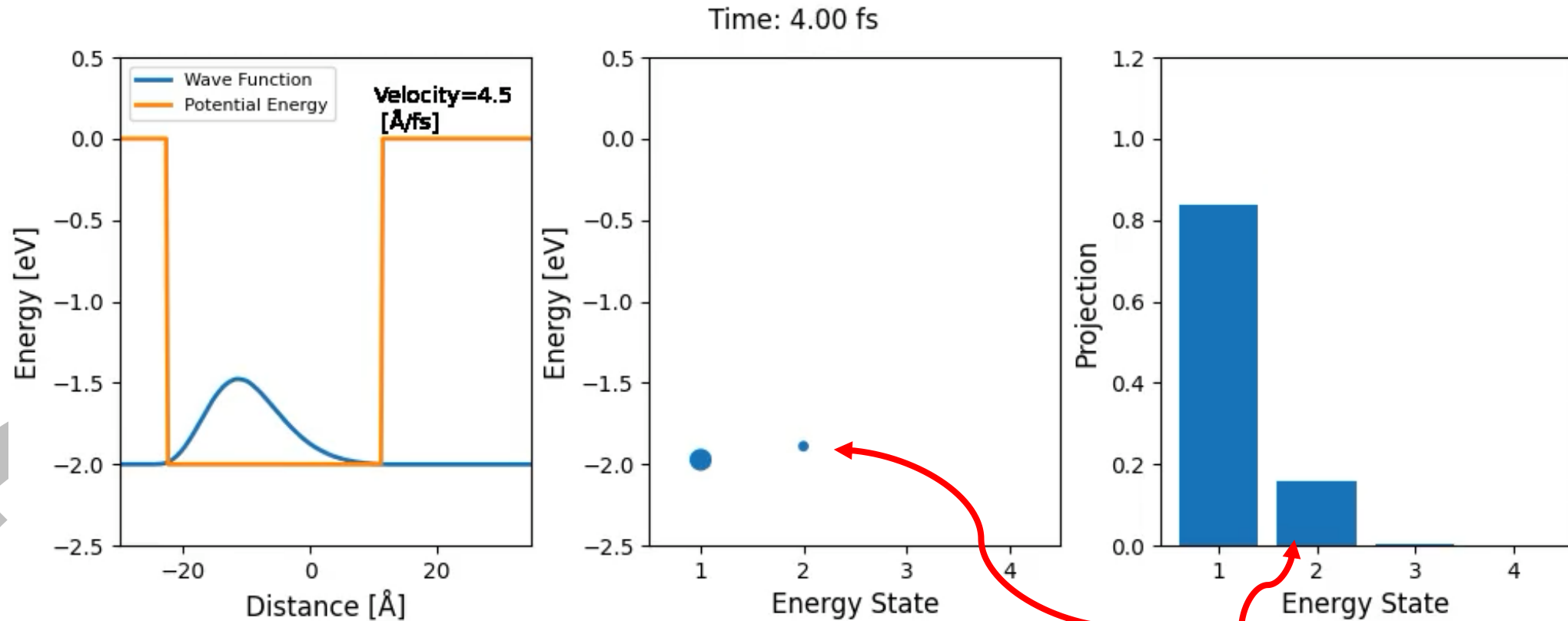


Case Two: Fast Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 4.5 Å/fs



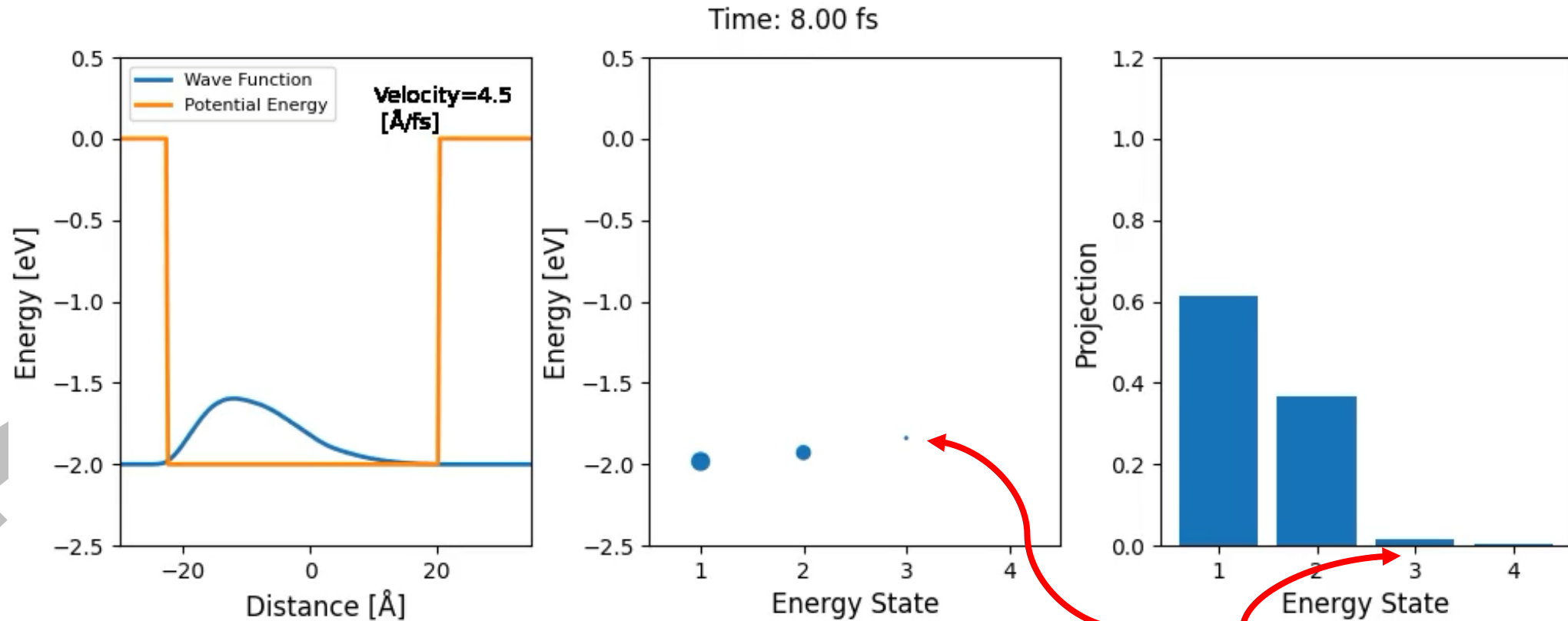
Excited state
contributions arise

Case Two: Fast Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 4.5 Å/fs



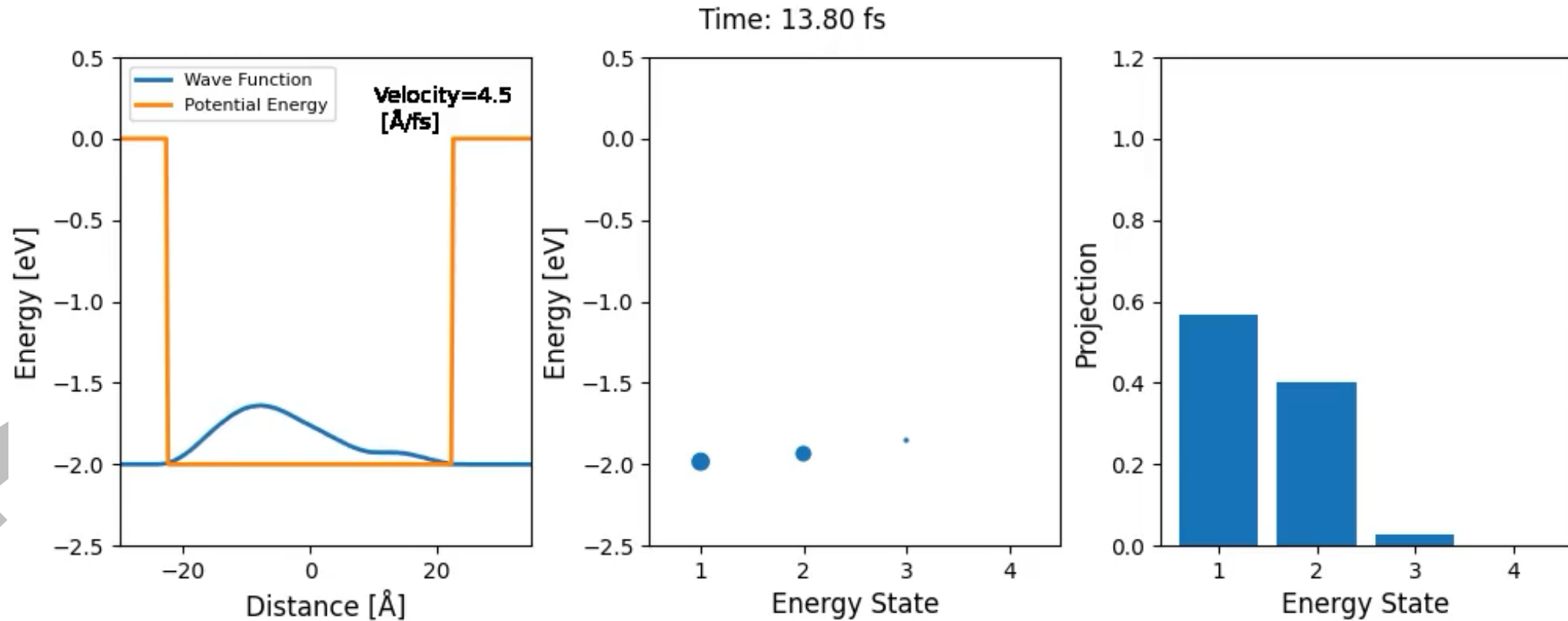
Second excited state
contributions arise

Case Two: Fast Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 4.5 Å/fs



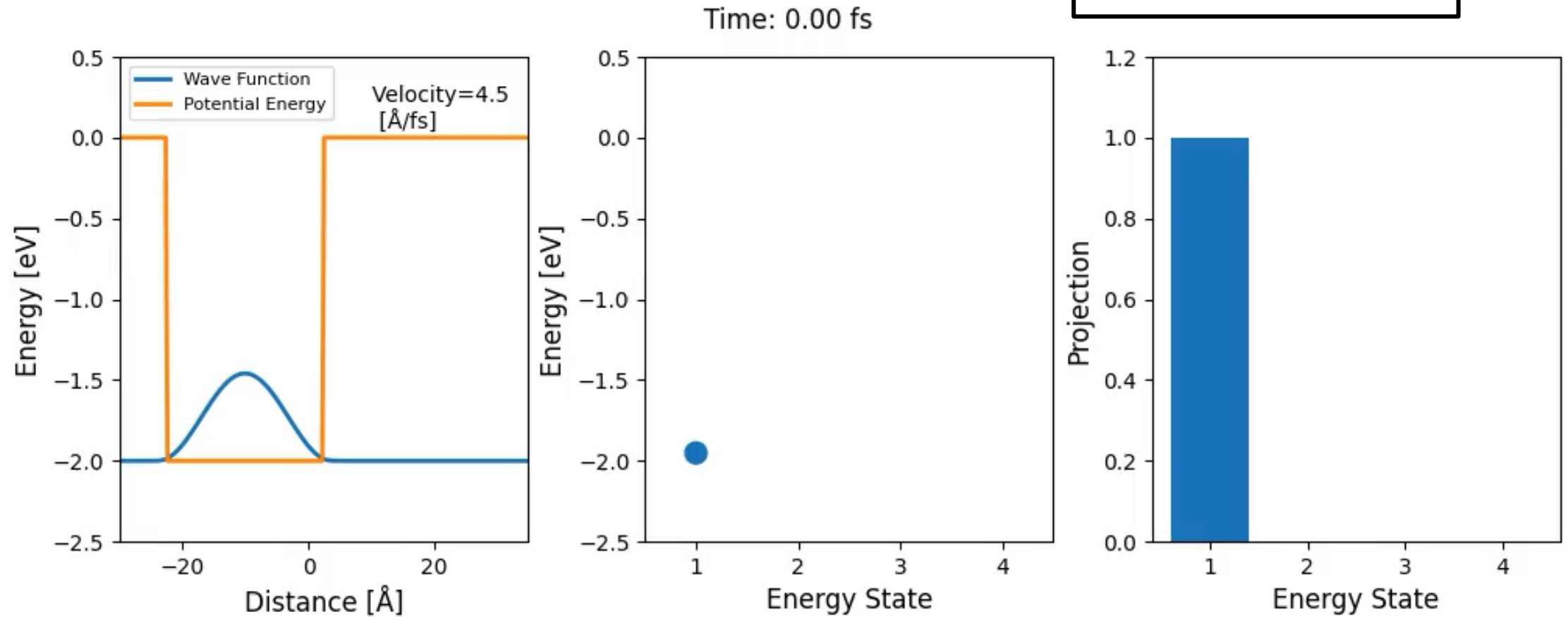
When expansion stops, the wave function tries to “catch up” to the larger box size

Case Two: Fast Expansion

Initial conditions

Well depth: 2 eV

Wall velocity: 4.5 Å/fs



The Glissando: Using sound as a qualitative descriptor



Remember:

1. Energy levels **decrease** as the box expands
2. Lower energies have **lower frequencies** ($E \propto f$)
3. Non-adiabatic dynamics excites and mixes wave functions into **higher energy** states

Sound 1

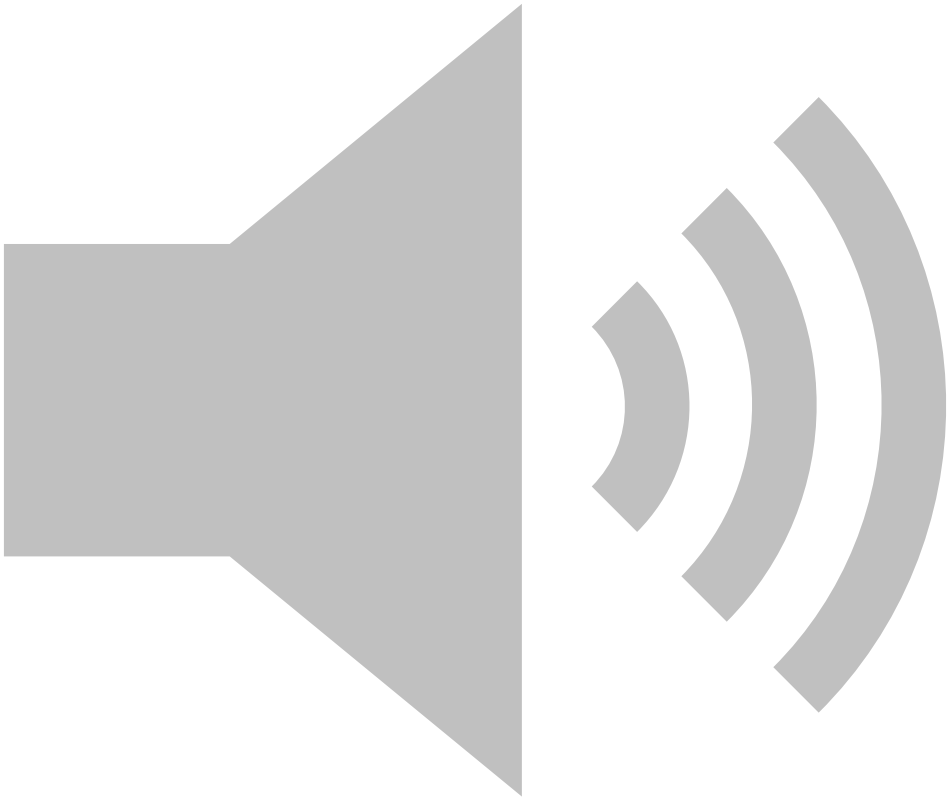


Sound 2

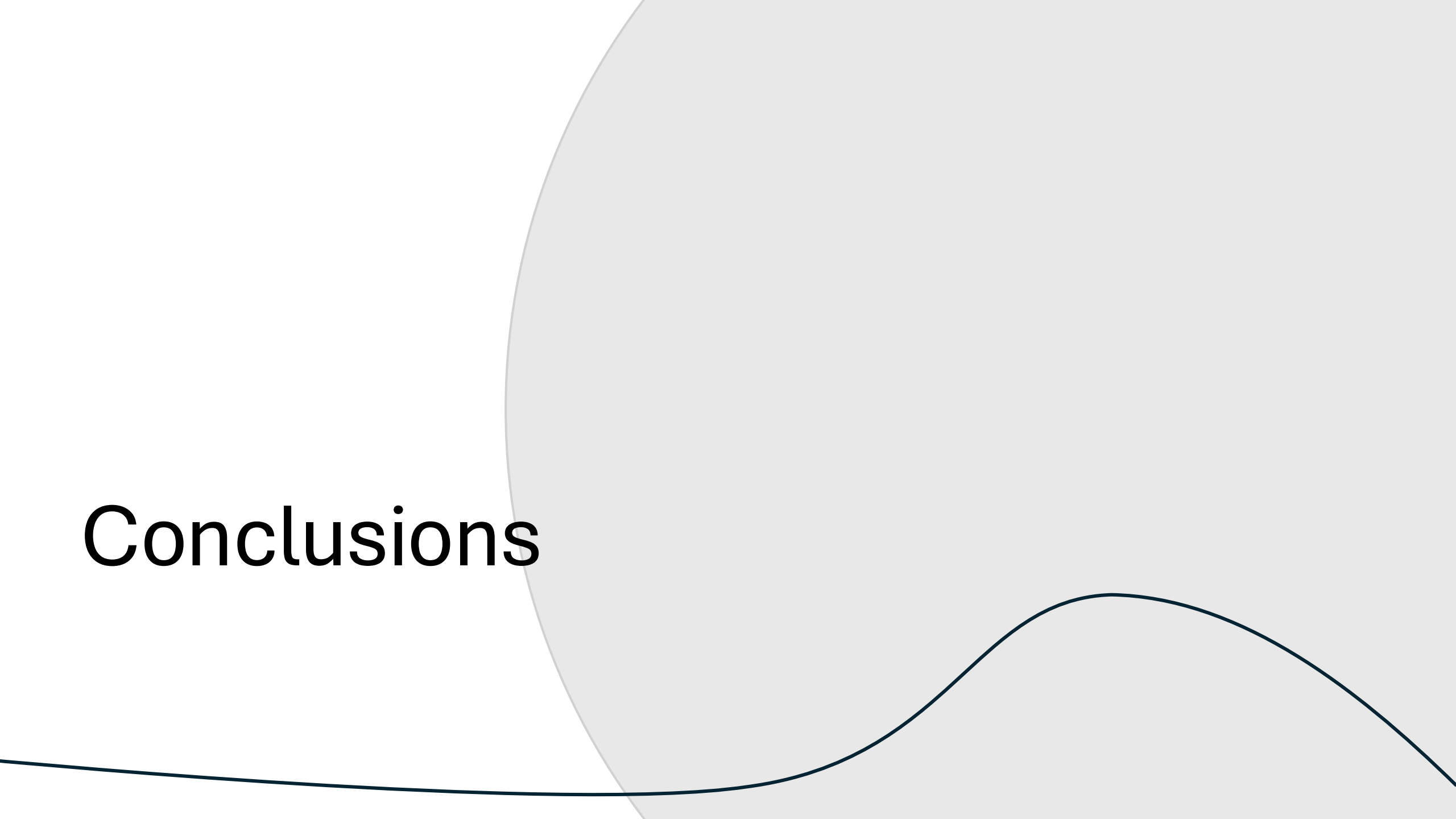


Using what we have discussed, which sound represents a fast-expanding box in the non-adiabatic regime?

ANSWER!

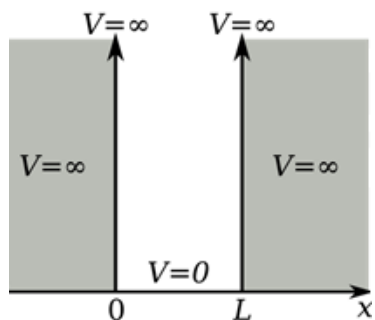


Sound 1
(non-adiabatic)

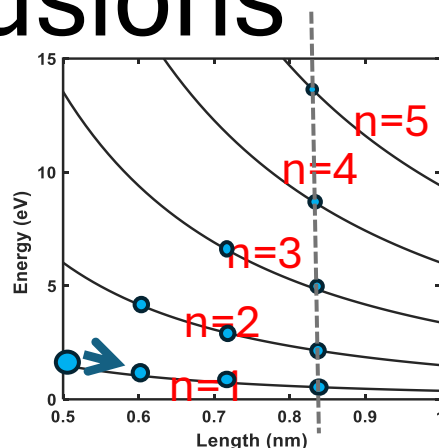


Conclusions

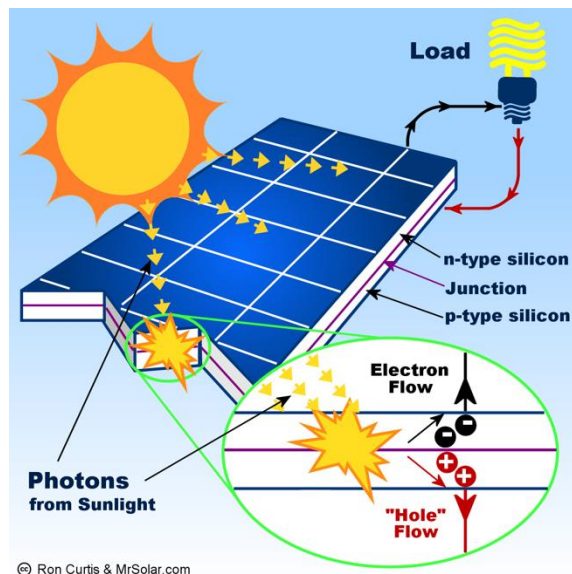
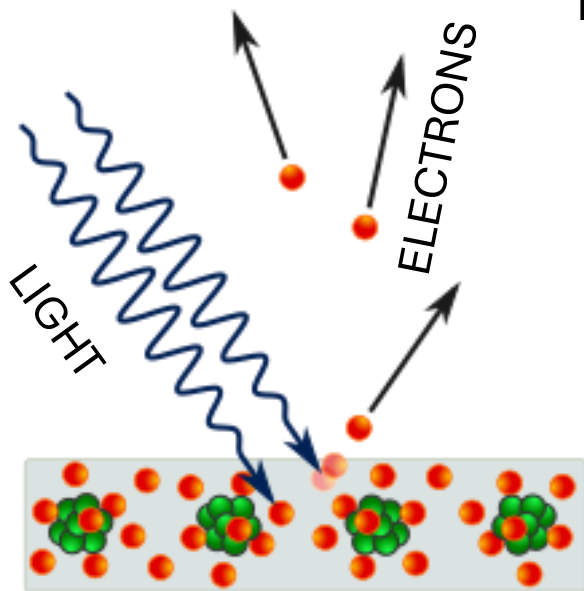
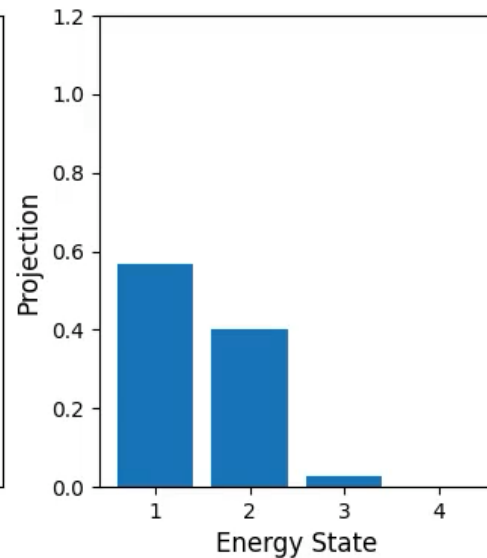
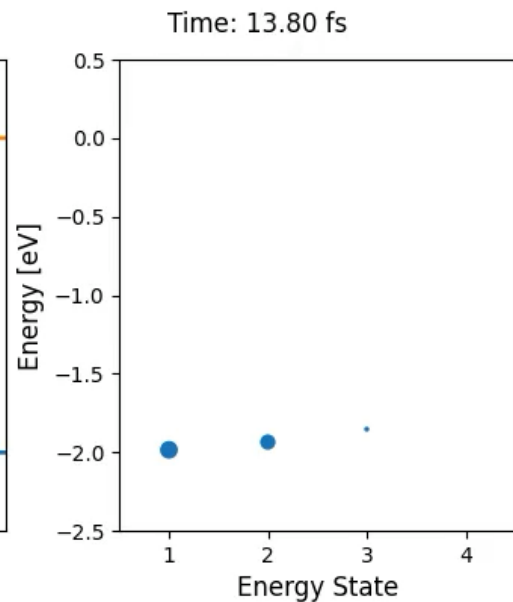
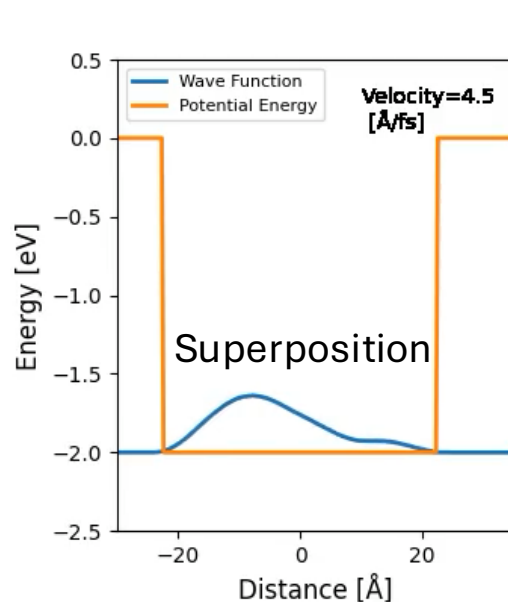
Conclusions



Quantum Mechanical Model



Excited States
Nonadiabatic
Energy Mixing



Real World Applications:
How can we apply our model to active research questions?

Other Research



Contents lists available at [ScienceDirect](#)

Artificial Intelligence Chemistry

journal homepage: www.journals.elsevier.com/artificial-intelligence-chemistry



Building a DFT+U machine learning interatomic potential for uranium dioxide

Elizabeth Stippell^{a,b}, Lorena Alzate-Vargas^a, Kashi N. Subedi^a, Roxanne M. Tutchton^a, Michael W.D. Cooper^c, Sergei Tretiak^{a,d}, Tammie Gibson^a, Richard A. Messerly^{a,*}

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Physical Insights into Materials and Molecular Properties | May 31, 2025 [Open Access](#)

Computational Screening of Ligands for Enhanced Interactions between Lead Halide Perovskite Quantum Dots

[Elizabeth Stippell](#); [Carlos Mora Perez](#); [Nicholas Favate](#); [Libai Huang](#); [Christina W. Li](#); [Oleg V. Prezhdo](#) *

+ [Author & Article Information](#)

J. Phys. Chem. Lett. (2025) 16 (23): 5666–5673.

<https://doi.org/10.1021/acs.jpcllett.5c01307> [Article history](#) ↻

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[Elizabeth Stippell](#); [Wei Li](#); [Claudio Quarti](#); [David Beljonne](#); [Oleg V. Prezhdo](#) *

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ACS Appl. Mater. Interfaces (2025) 17 (1): 2032–2040.

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PySyComp: A Symbolic Python Library for the Undergraduate Quantum Chemistry Course

Elizabeth Stippell, Alexey V. Akimov*, and Oleg V. Prezhdo*

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Supporting Information (3)

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Thank you!

This Work:

- Prof. Jahan Dawlaty (PhD Advisor, USC)

Other Collaborators & Mentors:

- Prof. Oleg Prezhdo (PhD Advisor, USC)
- Dr. Richard Messerly (LANL Mentor)
- Prof. Alexey Akimov (Undergraduate Research Advisor, UB)

Prof. Abraham Nitzan (Postdoctoral Advisor, Penn)

Extra Slides

From toy model to active research: What makes a good semiconductor?

- Calculating the rate of electron transfer provides key insights on semiconductor efficiency

Coupling influences
probability/coefficients

$$\Psi = \sum c_n \psi_n$$

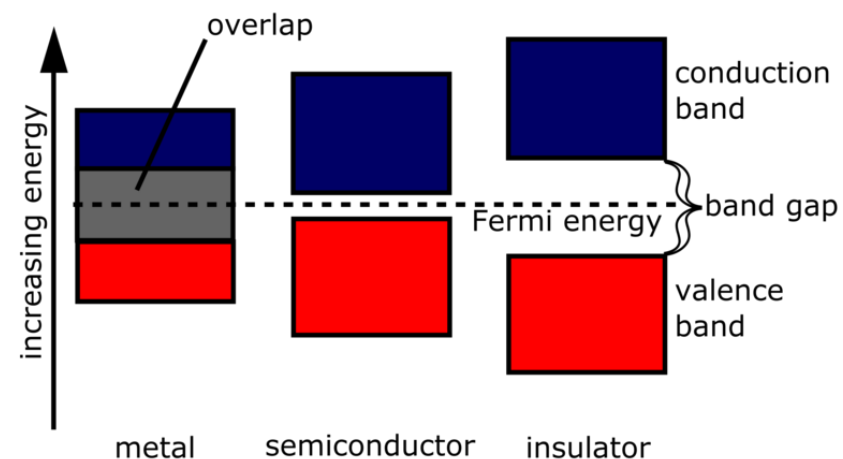
$$V_{kl} \propto c_n$$

Coupling
between energy
states!

Stability of the
material (energy
fluctuations)

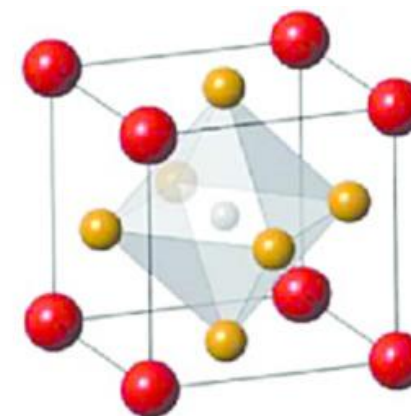
$$k_{\text{Marcus}} = \left(\frac{V_{kl}^2}{\hbar} \right) \sqrt{\frac{\pi}{\lambda k_B T}} \exp \left(-\frac{(\Delta A + \lambda)^2}{4\lambda k_B T} \right)$$

- As the coupling increases, the rate of electron transfer increases
- As the stability increases, the system's energy decreases and the rate increases



~22% efficiency

Large ion
Metal
Anion/Halide



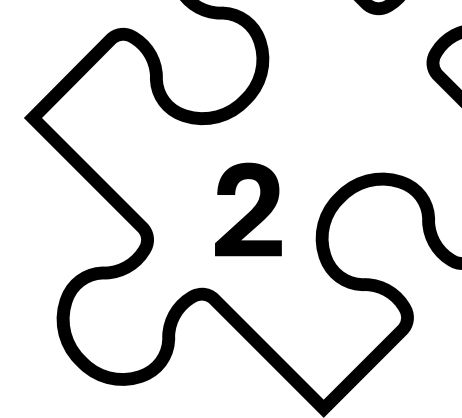
Learning Outcomes & Expectations

Develop basic programming skills through running (and debugging) Python, a free, open-source software

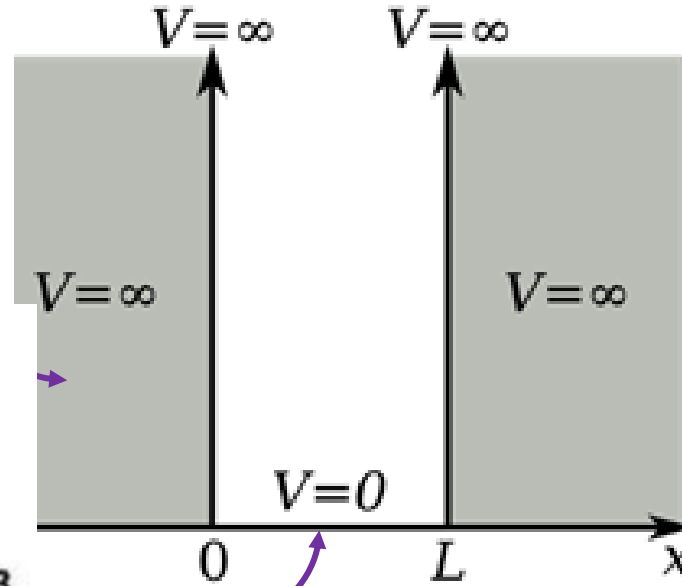
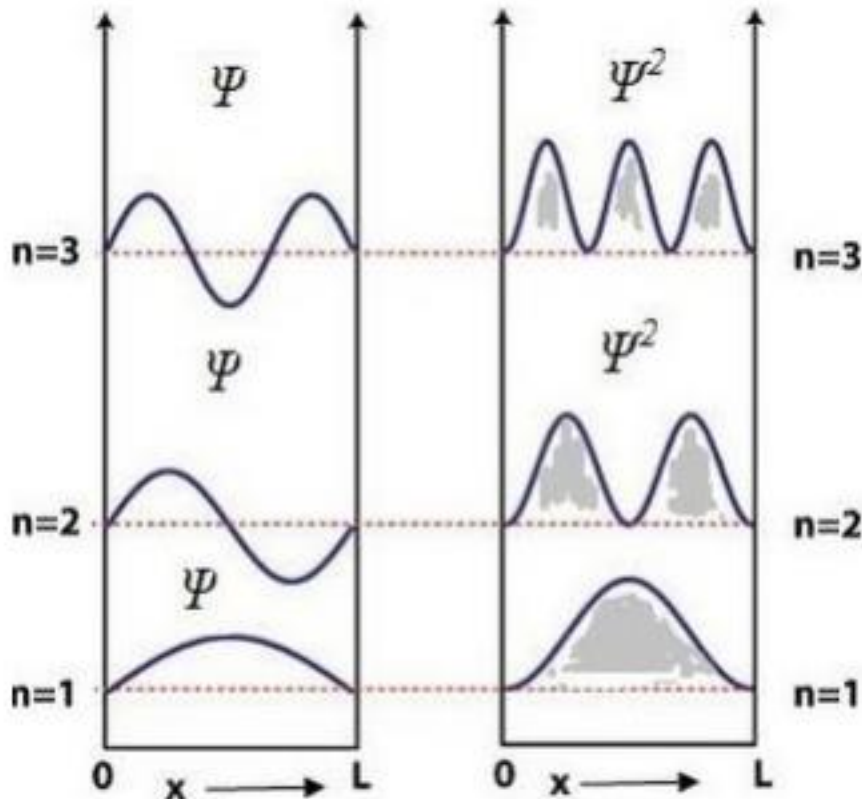
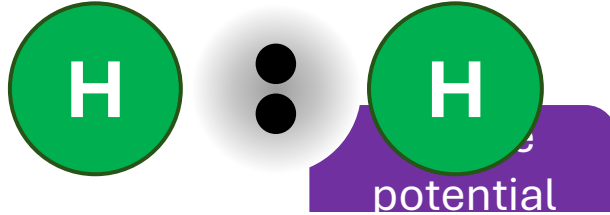
Understand non-adiabatic coupling, an important concept for real-world scenarios like in semiconductors and charge transport

Expand knowledge past the “basic” particle in a box model to show its capabilities for more advanced topics

The Particle-In-A-Box Model



Wave-Particle Duality



Zero
potential
inside



It's like red-light-green-light

- Show different energy states and their mixing (constructive and destructive waves)
- Superpositions are added together pieces.. Good example?? – keep adding sides to a shape?
- Discretized states?

How educational models can provide key insights into real-life problems

Particle in a box: conjugated carbon chains, quantum dots



Non-adiabatic coupling (NAC): applications in semiconductors

$$E \propto f$$
$$E \propto \frac{1}{L}$$

ACS Applied Materials & Interfaces > Vol 17/Issue 1 > Article

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FUNCTIONAL NANOSTRUCTURED MATERIALS (INCLUDING LOW-D CARBON) | December 16, 2024

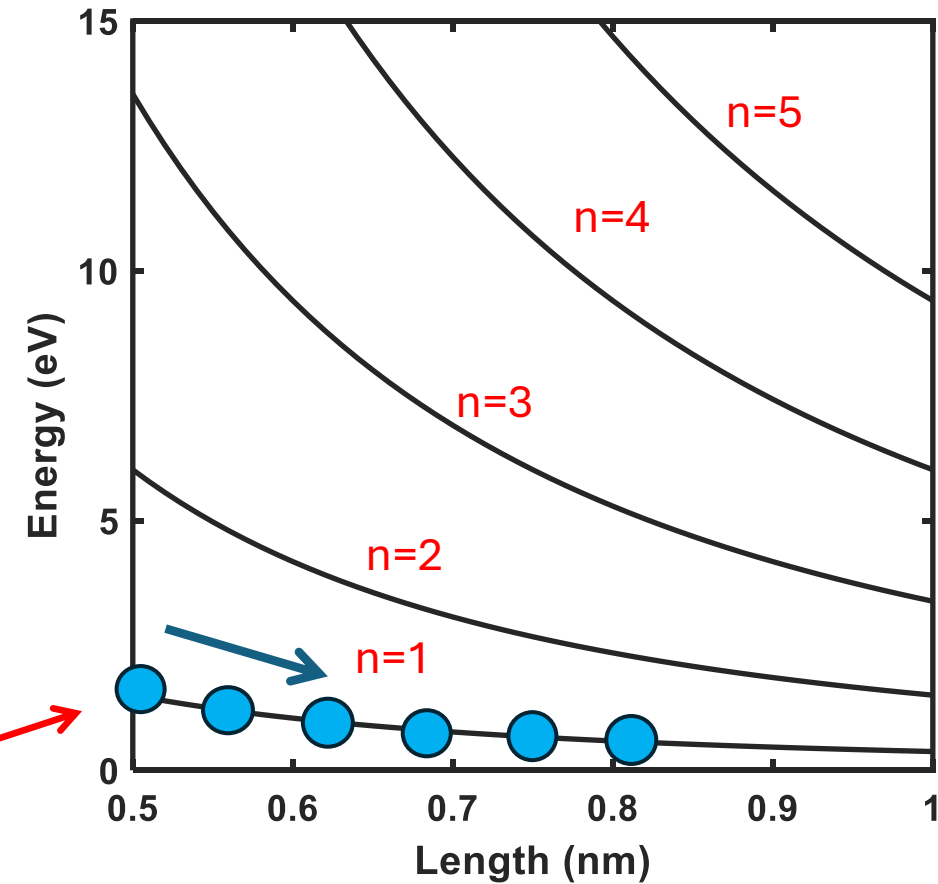
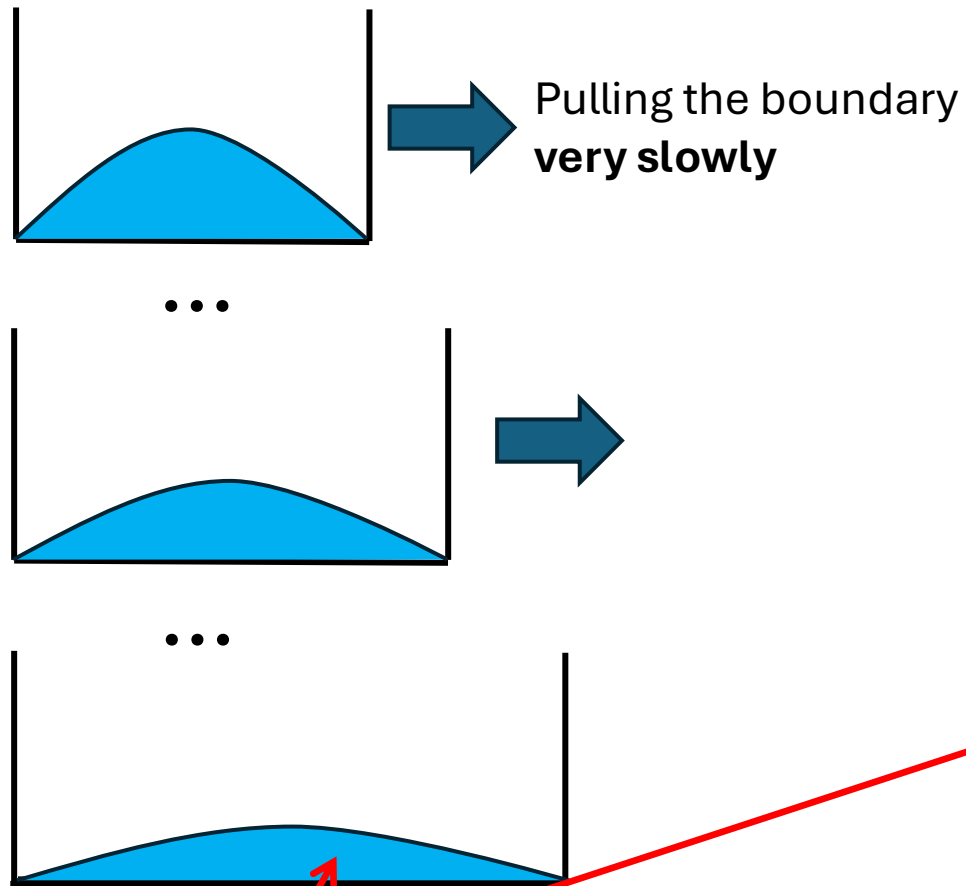
Enhancing Interlayer Charge Transport of Two-Dimensional Perovskites by Structural Stabilization via Fluorine Substitution

Elizabeth Stippell, Wei Li, Claudio Quarti, David Beljonne, and Oleg V. Prezhdo*

 Open PDF

Just talk about real-world PIB examples?

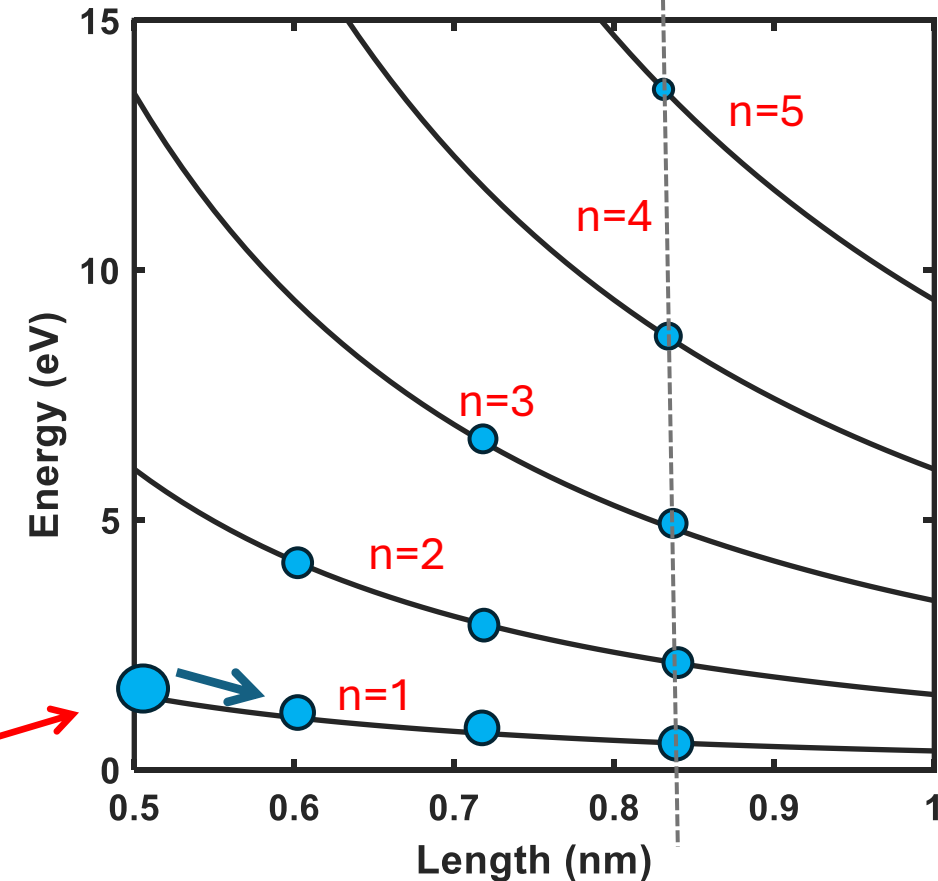
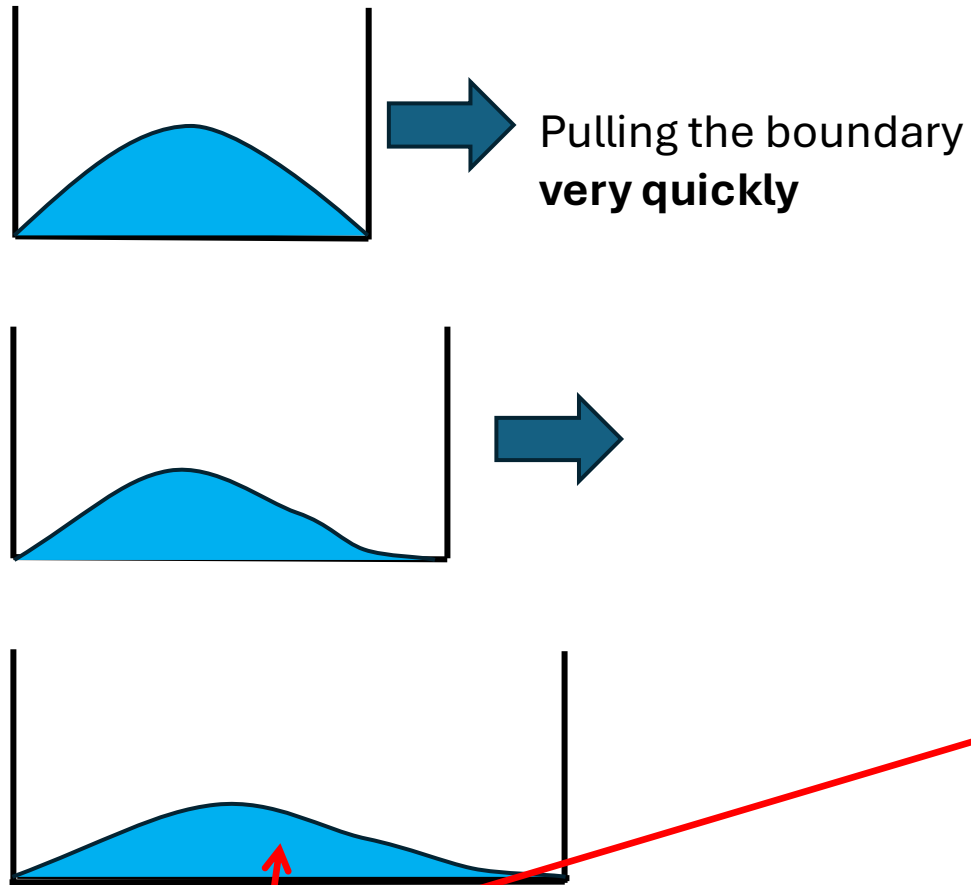
“Expanding” the particle in a box problem: **Slowly** Expanding Box



The particle remains in the ground state

The Adiabatic Limit.

“Expanding” the particle in a box problem: **Quickly** Expanding Box



The particle does not remain in the ground state and picks up character of the excited states during its evolution.

Non-adiabatic mixing of states.
The particle travels on several potential energy surfaces.

Non-Adiabatic Coupling

$$h_{jk} = \left\langle \Psi_j \left| \frac{\delta \hat{H}}{\delta \mathbf{R}} \right| \Psi_k \right\rangle$$

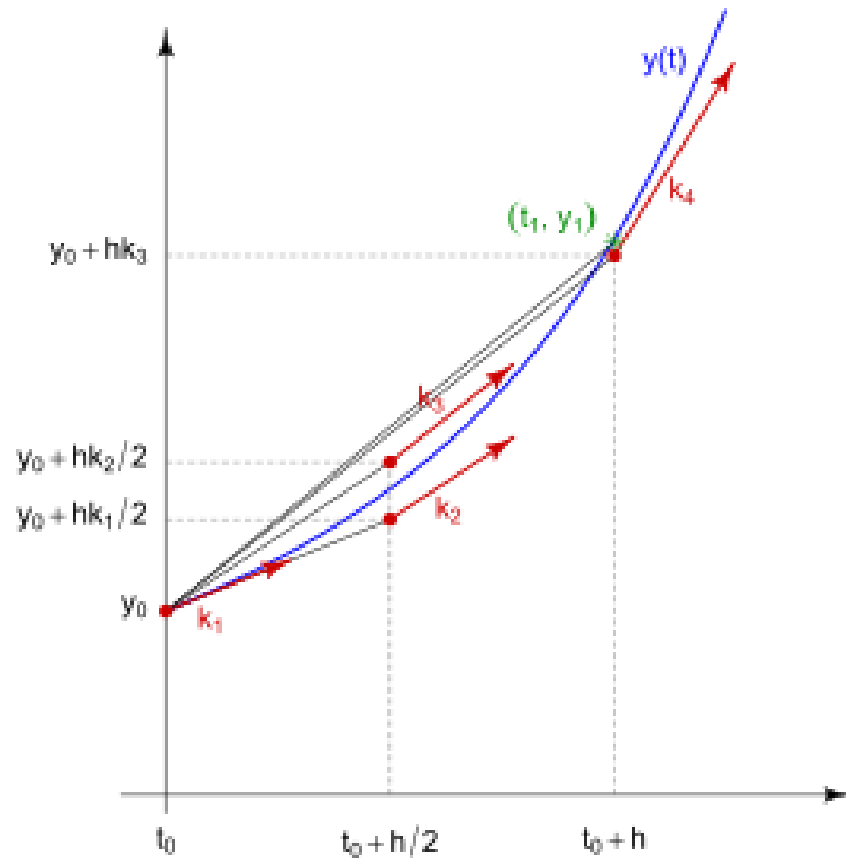
S.E.

Runge-Kutta 4th Order

Check paper

Summary of Methodology: Runge-Kutta

$$\Psi_{n+1} = \Psi_n + \frac{h}{6}(k_1 + 2 * k_2 + 2 * k_3 + k_4)$$



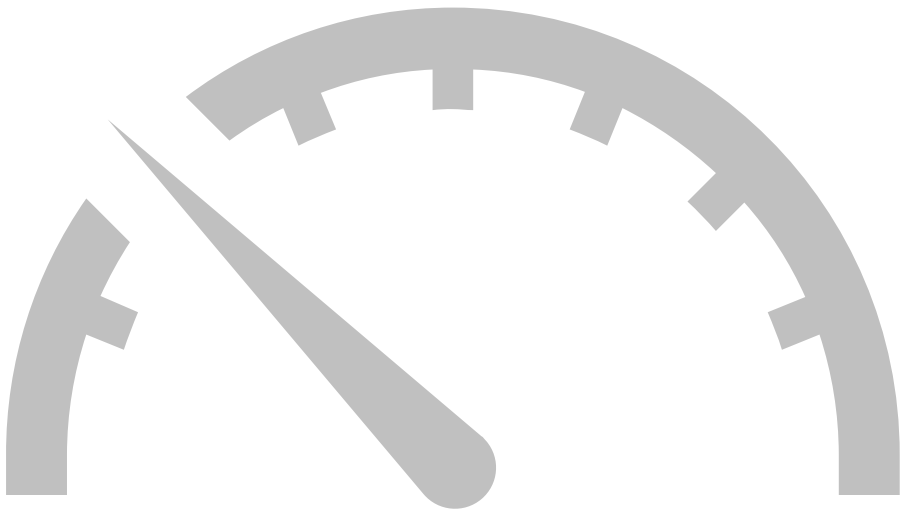
Runge-Kutta Methods.

A family of iterative methods which can be implemented to compute the wave function of our particle in a box.

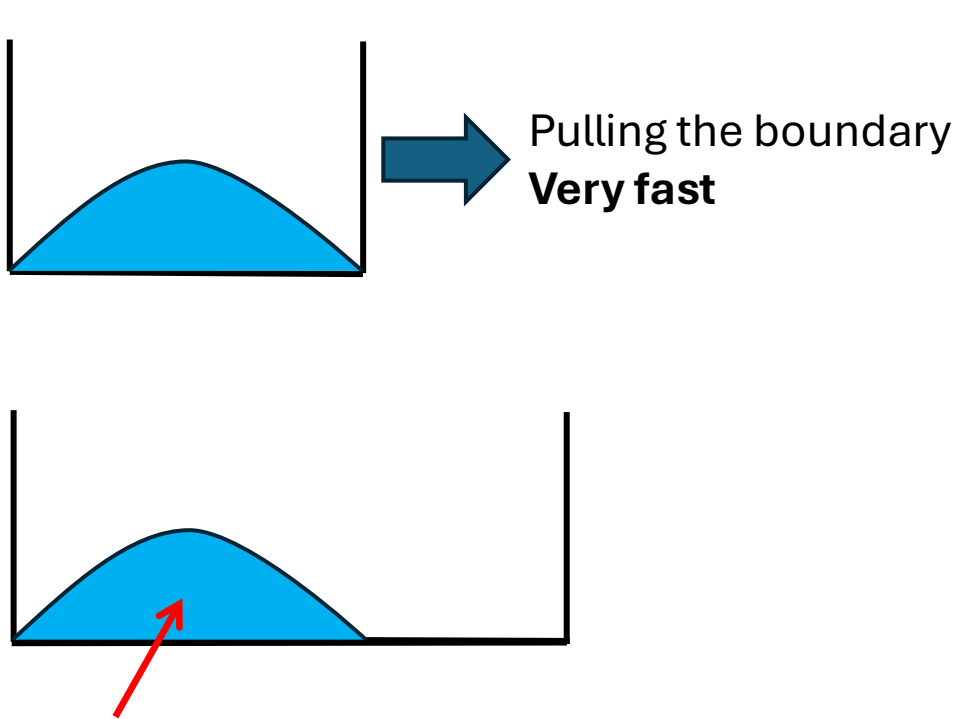
k_1	slope at start of interval
k_2	slope at midpoint
k_3	slope at midpoint
k_4	slope at end of interval

“Expanding” the particle in a box problem

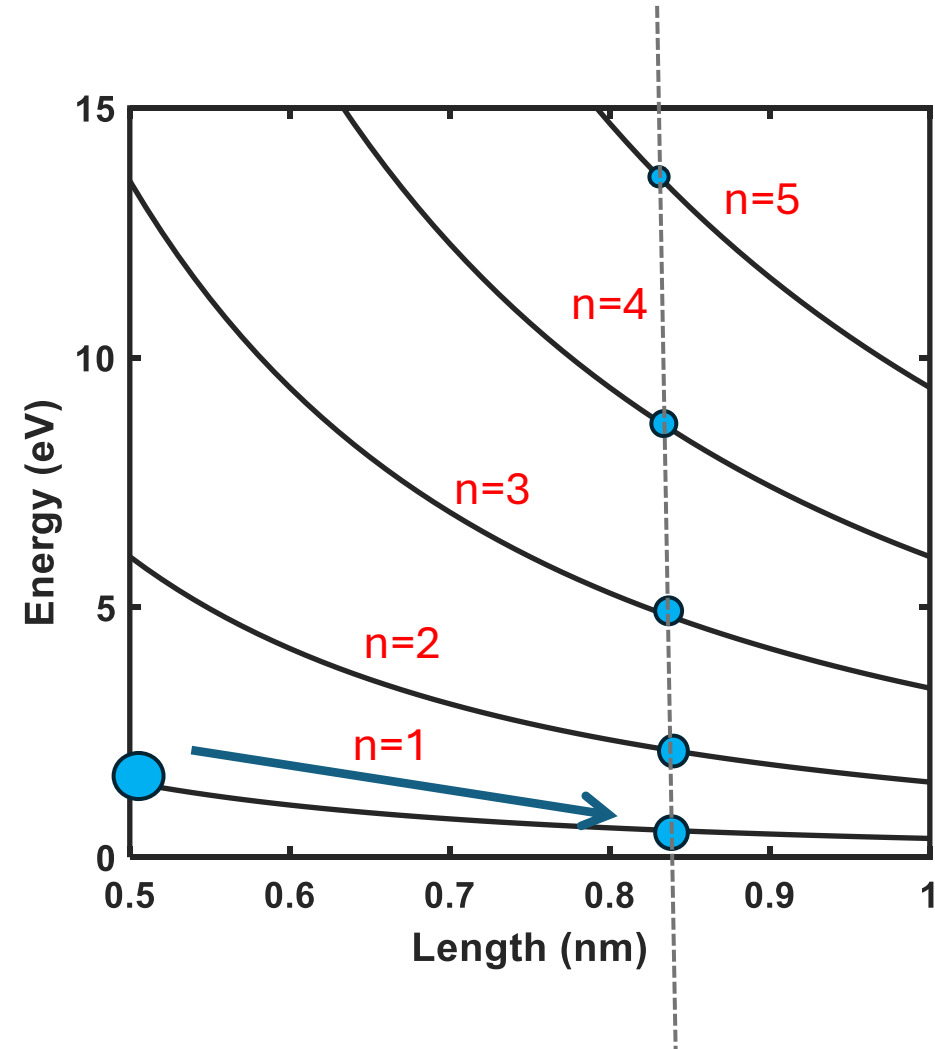
- Inducing NAC through rapid box expansion, therefore creating “Glissando”



Particle in an Infinitely Fast Expanding Box



The particle can not adjust to the sudden change in potential. Immediately after the change, the wavefunction looks like a superposition of many excited states. It evolves as a wave-packet afterwards.



The “Franck-Condon” Limit.

General statement about time evolution where eigenstates themselves are changing

Start

At the beginning, if we start from a single eigenstate

$$\Psi(0) = 1 \phi_1(0)$$

Coefficient of the ground state is 1.

The system starts in the ground state

Evolve

General State in the

Future

At any arbitrary time in the future, the state in its **most general** form can be expanded as:

$$\Psi(t) = \sum_n C_n(t) \phi_n(t) e^{i\theta_n(t)}$$

Time-dependent coefficients
(mixing beyond ground state can occur)

Time-dependent future eigenstates

Dynamic phase accumulation as the system evolves.

$$\theta_n(t) = -\frac{1}{\hbar} \int_0^t E_n(t') dt'$$

How do we find the coefficient that indicate mixing?

Coefficient time evolution

Mixture with other states is **DIRECTLY** proportional to how **FAST** the Hamiltonian is changing

$$\frac{d}{dt} c_j(t) = c_j(t) \left\langle \phi_j(t) \left| \frac{d}{dt} \phi_j(t) \right\rangle + \sum_{n \neq j} c_n(t) \frac{\left\langle \phi_j(t) \left| \frac{d}{dt} H(t) \right| \phi_n(t) \right\rangle}{E_n(t) - E_j(t)} e^{-i(\theta_n(t) - \theta_j(t))}$$

The rate of change
of the j^{th}
coefficient...

... is proportional to
the coefficient
itself...

... plus some extra
bits that depend on
other coefficients.

Mixture of state j with other states
INVERSELY depends on how
CLOSE they are in energy to state j .

Only results are shown. The derivations are found in many advanced quantum mechanics books.

The adiabatic approximation

$$\frac{d}{dt} \mathbf{C}_j(t) = \mathbf{C}_j(t) \left\langle \phi_j(t) \left| \frac{d}{dt} \phi_j(t) \right\rangle \right.$$

If the adiabatic approximation is adopted...

$$\mathbf{C}_1(t) = \mathbf{C}_1(0) e^{i\gamma(t)}$$

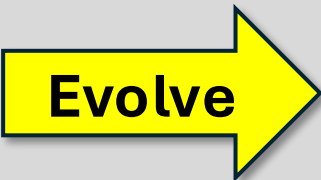
The initial coefficient never mixes with others...

$$\gamma(t) = i \int_0^t \left\langle \phi_j(t') \left| \frac{d}{dt'} \phi_j(t') \right\rangle dt' \right.$$

It just merely accumulates a phase
(known as Berry's phase)

If you **start in the ground state**

$$\Psi(0) = \mathbf{1} \phi_1(0)$$



You will **always stay in the ground state**, just picking up two phase factors.

$$\Psi(t) = \mathbf{1} \phi_n(t) e^{i\theta_n(t)} e^{i\gamma(t)}$$

1. Introduction

